

Luqi Sun

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M.S.E. Student in Electrical and Computer Engineering, Johns Hopkins University

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RESEARCH INTERESTS

- **Speech and Language Processing:** I am interested in applying speech and language processing methods to real-world healthcare problems, such as Alzheimer's disease and depression detection. I aim to understand how acoustic and linguistic cues reflect cognitive and mental health, and build robust models for reliable assessment across speakers and recording conditions.
- **Large Language Models (LLMs):** I am interested in improving the reasoning and generalization capabilities of large language models by studying their internal representations. My interests include neuron-level analysis and interventions to understand and control model behavior, as well as developing models that integrate speech and multimodal information.

EDUCATION

Johns Hopkins University , Baltimore, MD, USA	Aug 2025 – May 2027 (Expected)
Master of Science in Engineering in Electrical and Computer Engineering	GPA: 3.47/4.0
Dalian University of Technology , Dalian, China	Sep 2021 – Jun 2025
Bachelor's Degree in Digital Media Technology	Major GPA: 88.49/100

RESEARCH EXPERIENCE

Research Assistant, Johns Hopkins University SMILE Lab , Center for Language and Speech Processing (CLSP) Advisor: Prof. Berrak Sisman	Aug 2025 – Present
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- Develop robust models for healthcare applications using speech and language processing methods, including Alzheimer's disease and depression detection. Investigate acoustic and linguistic cues associated with these conditions and evaluate model reliability across datasets and recording conditions.
- Investigate neuron-level interventions in large language models (LLMs) to understand and control their behavior, with an emphasis on how internal representations influence model outputs. Contribute to LLM development through model experimentation and evaluation of language understanding and generation capabilities.

Research Assistant, Dalian University of Technology Advisor: Prof. Qiufen Xia	Feb 2024 – May 2025
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- Investigated cold-start latency in serverless computing and proposed a nested architecture to mitigate this issue. This work resulted in a first-author paper and a granted patent.

PUBLICATIONS

- [1] **Luqi Sun**, Shreeram Suresh Chandra, Lin Zhang, You-Jin Li, Brian MacWhinney, Yu Tsao, Emily Mower Provost, and Berrak Sisman. [Cleaner Speech, Weaker Generalization: Revisiting Pitt-Derived Benchmarks for Alzheimer's Disease Detection](#). *AAACL-IJCNLP Main Conference*, 2026.
- [2] Xiutian Zhao, **Luqi Sun**, Björn Schuller, and Berrak Sisman. [Emotion Across Speech and Faces: Shared Affective Mechanisms in Multimodal Foundation Models](#). *IEEE Spoken Language Technology Workshop (SLT)*, 2026.
- [3] Hsiang-Chen Yeh, **Luqi Sun**, Aurosweta Mahapatra, Shreeram Suresh Chandra, Emily Mower Provost, and Berrak Sisman. [Who is Speaking or Who is Depressed? A Controlled Study of Speaker Leakage in Speech-Based Depression Detection](#). *Interspeech*, 2026.
- [4] [Under Review] **Luqi Sun**, Shreeram Suresh Chandra, Aurosweta Mahapatra, Emily Mower Provost, Brian MacWhinney, and Berrak Sisman. [Robust Cross-Domain Speech-Based Alzheimer's Disease Detection via Iterative Adversarial Self-Training](#). *Submitted to ICASSP*, 2026.
- [5] **Luqi Sun**, Qiufen Xia, Xiaolong Zhai, Zhen Feng, and Jiankang Ren. [The Optimization of Container Cold Start Latency in Serverless Computing by Nested Architecture](#). *21st International Conference on Mobility, Sensing and Networking (MSN)*, 2025.

PATENT

Serverless Computing Container Cold Start Method and Related Device Based on Nested Architecture. Chinese Patent **CN120821531B**.